



8/13/2026

ASSIGNMENT DAY2

Hardware Failure Diagnosis & Multimeter Testing.

TASK 1.

❖ List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' - symptom: blank screen, no POST beep).

➤ Common Desktop PC Hardware Issues:

<u>Hardware Issue</u>	<u>Typical Symptom</u>
No Display	Blank screen, no display output
Faulty RAM	PC fails to boot or crashes frequently
Power Supply Failure	PC does not turn on
Hard Disk/SSD Failure	Slow performance or boot failure
CPU Overheating	Sudden shutdowns or system freezing

TASK 2.

❖ Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

➤ Laptop Not Charging, Troubleshooting Plan:

🔧 **Check the Power Outlet:** Test another device to make sure the outlet is working.

- ✚ **Inspect the Charger:** Check the adapter and cable for cuts, damage, or loose connections.
- ✚ **Check the Charging Port:** Look for dust, dirt, or a loose connection in the laptop port.
- ✚ **Test with a Multimeter:** Check the charger's output voltage to confirm it is working properly.
- ✚ **Check the Battery:** If the charger works but the laptop still does not charge, the battery may be faulty.

TASK 3.

- ❖ **Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).**

➤ USB Charger Voltage Test :

✚ **Measured Voltage:** 5.02 V

✚ **Expected Range:** 4.75 V – 5.25 V

✚ **Result:** The measured voltage is within the expected range, indicating that the charger output is normal.

TASK 4.

❖ A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate? Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

➤ Testing the 12V Rail of an ATX Power Supply:

- Set the multimeter to DC voltage (V —) mode.
- Connect the probes: red probe to the **yellow wire (+12V)** and black probe to the **black wire (GND)**.
- Record the reading: **12.08 V**.
- **Result:** The voltage is within the normal range. A reading below **11.5 V** may indicate a **weak or faulty power supply**, which can cause random shutdowns.